

Element F: Consideration of Design Viability

Solution

- The design will be created with a wooden block to provide strength and stability and metal made from recycled materials to provide the balance at the bottom. This book end will be carved and decorated to be aesthetically pleasing to appeal to students and staff.

Structurally Sound - Solution

- A requirement from Mrs. Tatro about the bookend was that it needed to be capable of supporting the weight of books and would last a long time in the library. Our proposed solution, to meet these requirements, needs to be structurally sound. The solution will be made using wood and metal. These 2 materials are known for their incredible strength and long life spans. As such, the solution will be structurally sound and satisfy the requirements of our stakeholders.

Solution Review for Viability

- It may be difficult to create a lightweight design
 - This obstacle can be realistically addressed by maintaining constant communication with our stakeholders to ensure that the design is a comfortable and convenient weight. This is viable because Mrs. Tatro emphasized that she wanted the bookends to be easy to move and not too heavy.
- It could require extra time and planning to test our design and make necessary changes
 - This obstacle can be realistically addressed by mapping out our schedule with margin for error and extra testing days to avoid missing deadlines. This is viable because Mr. Eisenburg suggested that we map out our deadlines well and don't spend too much time and don't rush.
- It could be difficult to create a long-lasting design
 - We will do research on each of the materials we are considering to determine which will hold up the best. This should work because Mrs. Tatro said that she wanted a long-lasting design that would not fall apart quickly.

Satisfaction

- As mentioned, we will constantly communicate with our stakeholders as we progress to ensure their approval at all times
- We are aware of the stakeholders' expectations and have sketched a design that will address more than one: a functional book end that is also decorative

Design Viability Description	Connection to Design Requirement	Credible Evidence?
The solution is inexpensive	All materials are refurbished	Wood scrap, borrowed sandpaper, refurbished hangers, and resin
Uses recyclable materials	Aligns with sustainability goals	All recyclable and eco friendly materials
Base is stable enough to support books	Stability rating at least $\frac{3}{4}$ in testing	Will test design with 5 hardcover books
Visually appealing	Meets needed design aesthetics	Will paint and use resin to customize